Fraud Document Detection Tool

(FDDT)

User Manual

A guide for submitters, Reviewer L1 / Reviewer L2 and platform administrators

Describes the application as implemented — companion to the technical documentation in docs/

This manual is direct about what the automated checks can and cannot prove. No signature is ever "verified," AI-generated-content detection is experimental, and every scored case still goes to a human reviewer — nothing here auto-approves anything.

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FDDT User Manual

Fraud Document Detection Tool — a guide for submitters, reviewers and platform administrators.

This manual describes the application **as implemented** (see `docs/README.md` for the technical documentation this is derived from). It covers what each of the four roles can do, how a case moves from upload to decision, how to read the automated checks and the risk score, and — deliberately — what the tool does **not** do, so nobody relies on it for more than it promises.

> **One deployment, several companies.** The application serves several client companies at once. Every user belongs to exactly one company and only ever sees that company's cases, documents, settings and audit log. The platform operator's own staff (platform admins) sit outside every company.
>
> *Screenshots were captured before multi-company support was added; some labels (for example "Admin") are older than the text, but the screens work as described here.*

Who this is for

Role	Can do
User (submitter)	Submit a case with supporting documents, track its status, see the reviewer's decision. Never sees the risk score, tier or triggered reasons — only a neutral "In review" flag — so the checks that catch fraud are never exposed to the person submitting the documents.
Reviewer L1	Everything a user can do, plus: the full review queue and dashboard, every case's automated findings and risk score, approve / reject / escalate, audit history, and exporting the per-case forensic report. Once a case is escalated to L2, a Reviewer L1 can still open and read it but can no longer approve, reject or re-escalate it.
Reviewer L2	Everything a Reviewer L1 can do, plus: approve or reject cases that have been escalated to L2, and manage **your company's** Issuer Registry and Risk Rules (Settings).
Platform Admin	Belongs to no company. Creates companies and every user account, edits the default rule set new companies start from, sees billing & usage per company and the processing-queue monitor. Can **read** any company's cases, audit log and settings for support — every such view is recorded in a platform-only log — but can never approve, reject, escalate, upload, create cases or export reports.

Every rule above is enforced by the backend on every request, not just hidden in the interface — reaching a screen you're not permitted to use still returns "access required" or a 404, never data. A case or document of another company simply does not exist for you (404).

Signing in

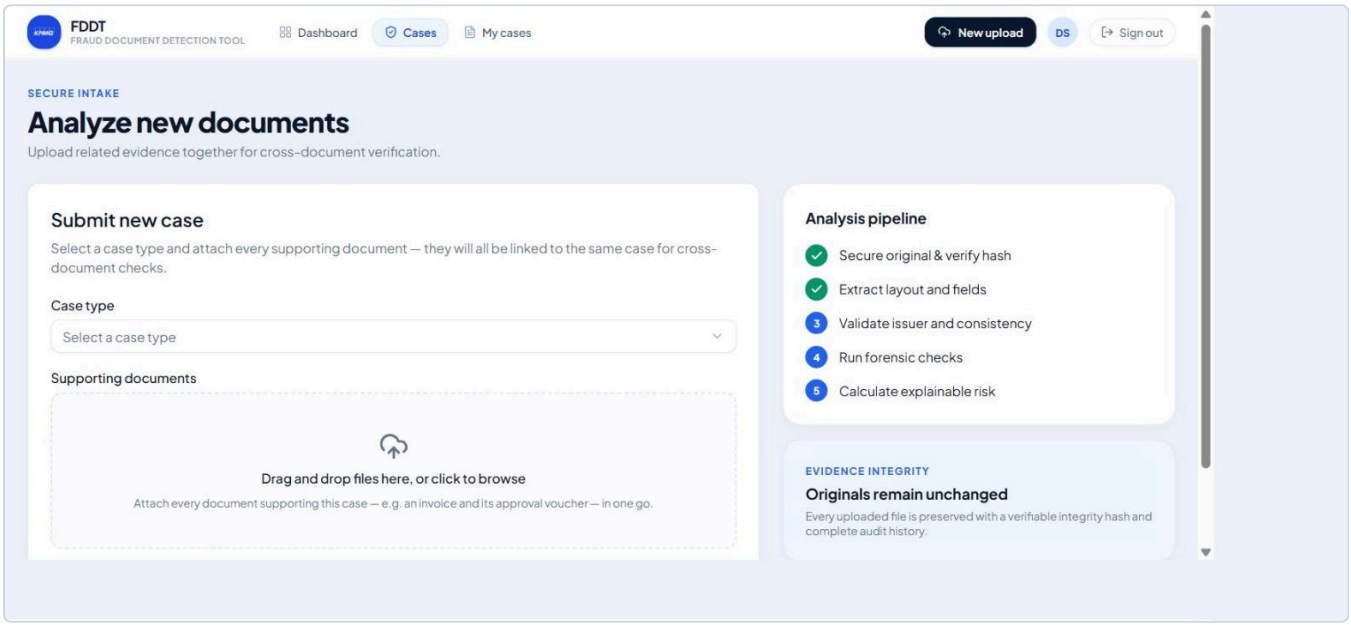
Go to the application URL and sign in with the email and password a platform admin created for you (there is no self-service sign-up, and no password-reset link — if you're locked out, ask the platform operator). The top bar shows your company and role. Sessions are a plain bearer token (8 hours) with no server-side logout. If your company is suspended, sign-in stops working for everyone in it.

> The sign-in screen also shows a "Trust this device" checkbox and a "Sign in with SAML / Okta SSO" button. Neither does anything: the checkbox isn't wired to the login form (authentication doesn't have a device-trust concept to back it), and the SSO button is disabled. Treat the screen as plain email/password sign-in only — the badges and extra controls are decorative UI that hasn't been wired up or removed yet.

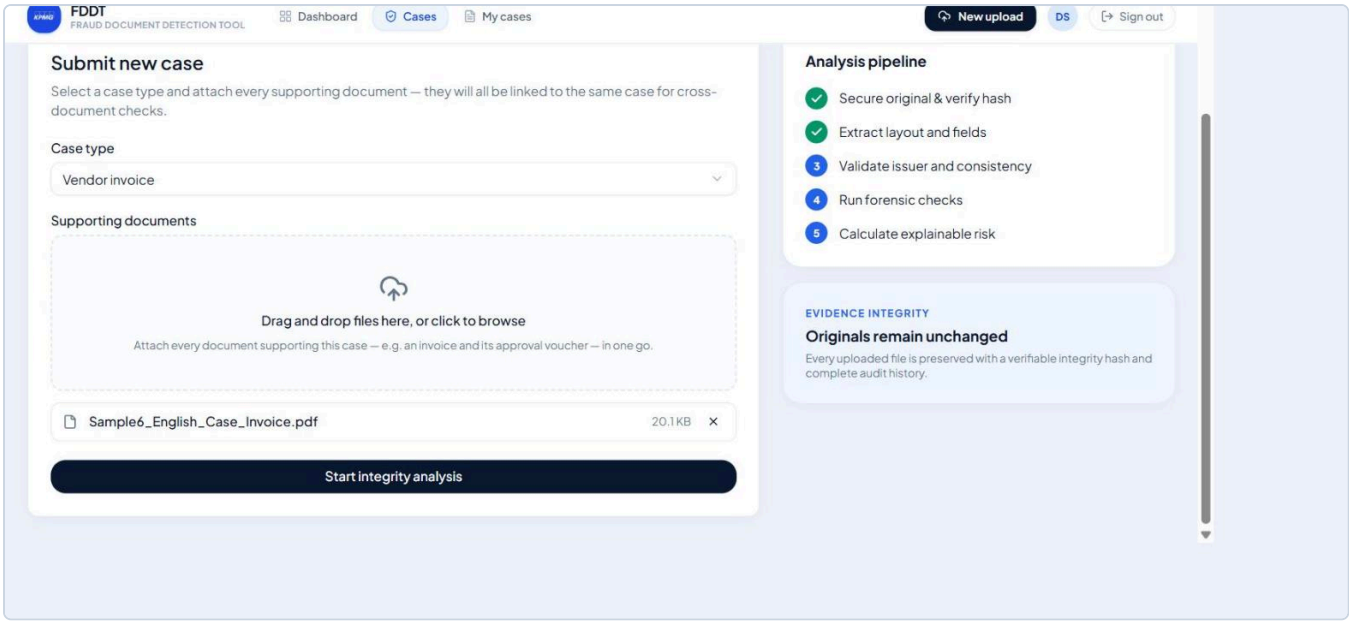
Submitting a case (User)

1. Open **New Case**.
2. Choose the case type: *School / educational document*, *Vendor invoice*, *Commercial invoice*, *Procurement documentation*, *Quotation*, *Travel / accommodation reimbursement*, or *Other*.
3. Attach the supporting document(s) — the original invoice, receipt, quotation, evidence of payment, etc. **Only PDF files are accepted, up to 10 MB each.** If you have a photo or scan as an image, save or scan it to PDF first. A file is refused immediately, with the reason shown next to it, if it is:
   - not a PDF (for example a JPEG, PNG or Word file, even if renamed to `.pdf`);
   - empty, or larger than 10 MB (the message shows the file's size and the limit);
   - **password-protected** — remove the password and upload it again (the system never asks for document passwords);
   - corrupted or cut off (for example a partly downloaded file).
4. Upload. Each file uploads independently with its own progress bar; if one fails you can fix it and retry without losing the others, remove the rejected file by clicking its `[X]` icon to proceed with the remaining documents, or click **Reset form** to clear the intake form and start fresh.
5. **Optional — set a reference signature:** for an uploaded document, you can draw a box around a signature or stamp so the system compares it against the one on other documents. Skip this and no signature comparison runs for that document.
6. The case now moves into the automated pipeline, and you're taken to its detail page.

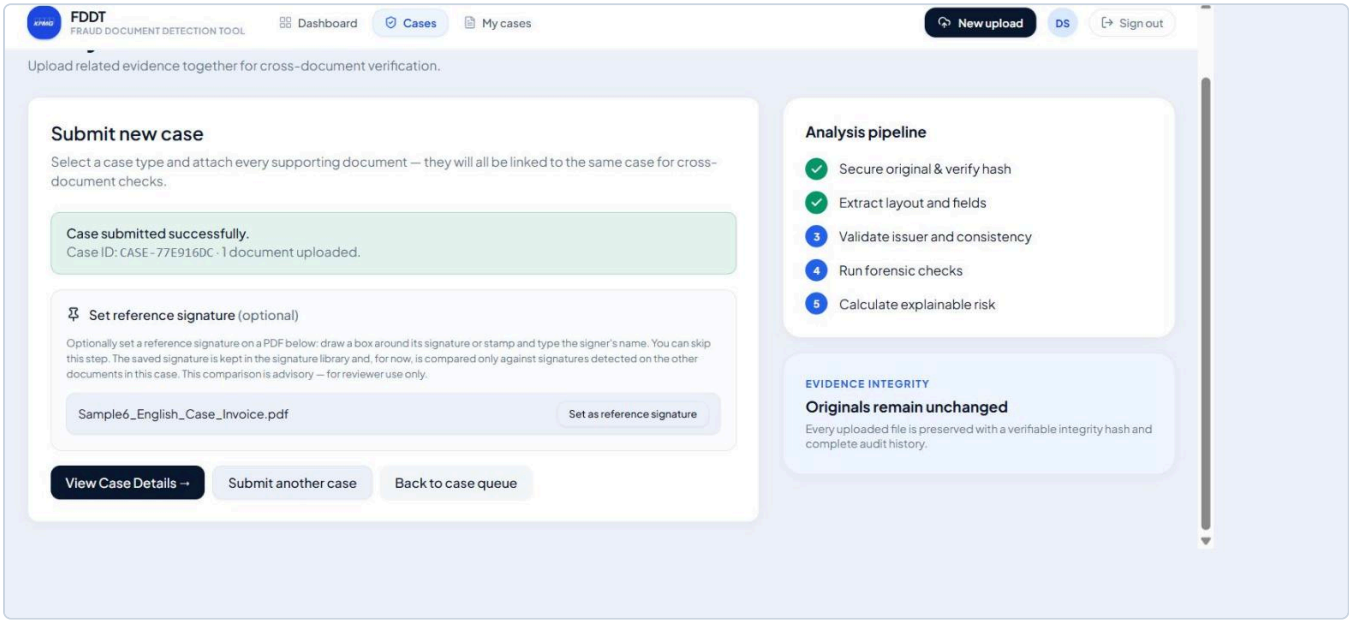
Attach every document of a case in this one submission — there is no "add a document" button on an existing case.



The New Case form



A case type chosen and a document attached, ready to submit



Confirmation after submission, with the optional reference-signature step

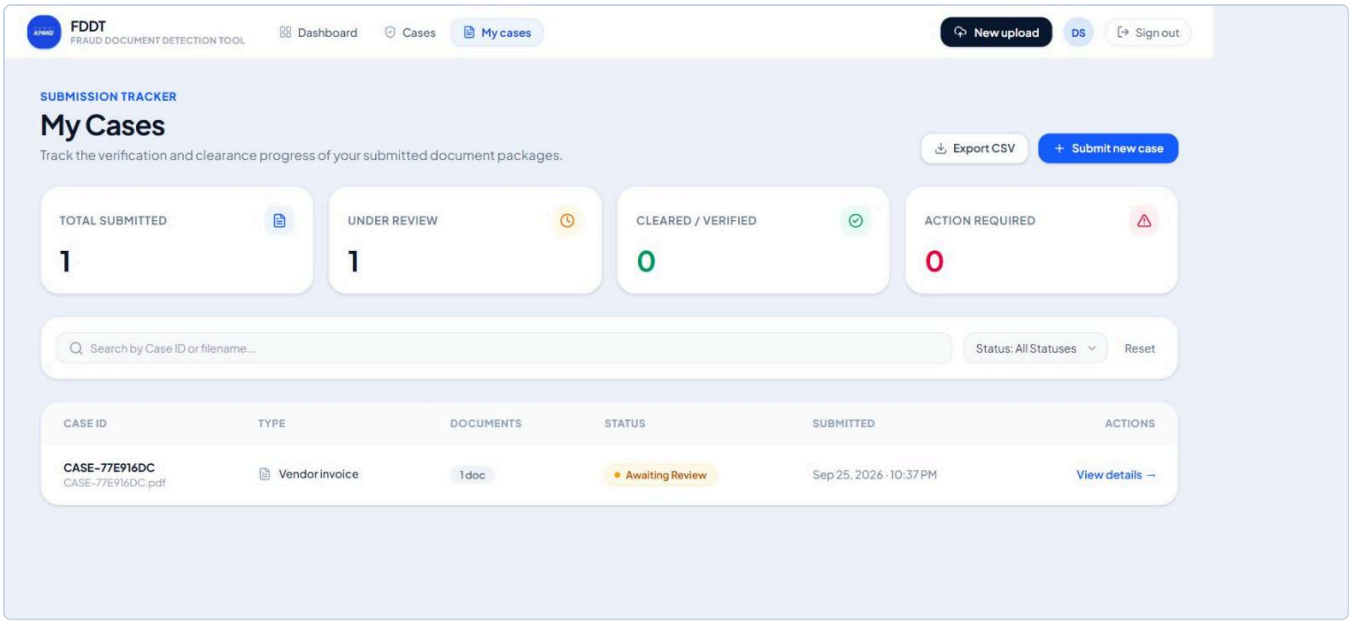
Tracking your cases (User)

My Cases lists everything you've submitted, filterable by *Under review*, *Cleared* or *Action required*, with search. You'll see the outcome (approved / rejected, with the reviewer's reason when rejected) but not the risk detail behind it — that's intentional (see below).

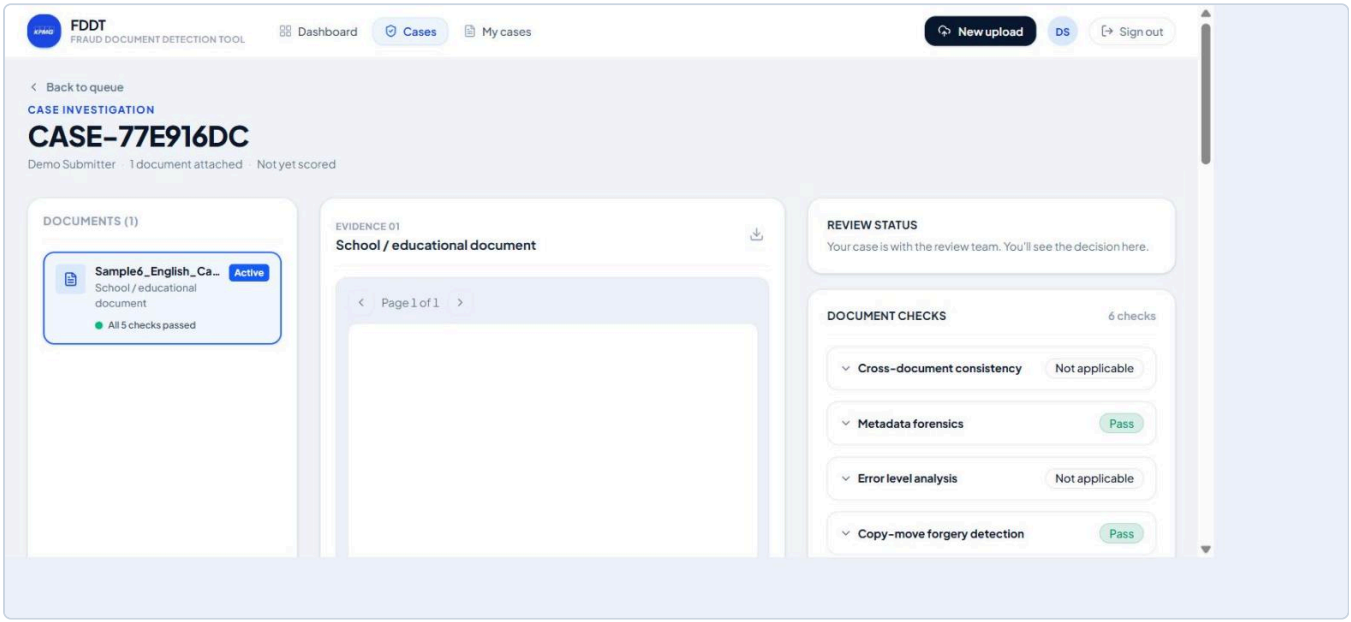
Opening one of your cases shows its documents, their extracted fields, and an activity timeline, but:

- No risk score, tier or list of triggered findings.
- No reviewer decision controls beyond a read-only history of what happened.

This is deliberate: if a submitter could see exactly which signals fired, a bad actor could learn what to avoid next time.



My Cases: a submitter's own cases and their status



A submitter's view of their own case — no risk score or tier, by design

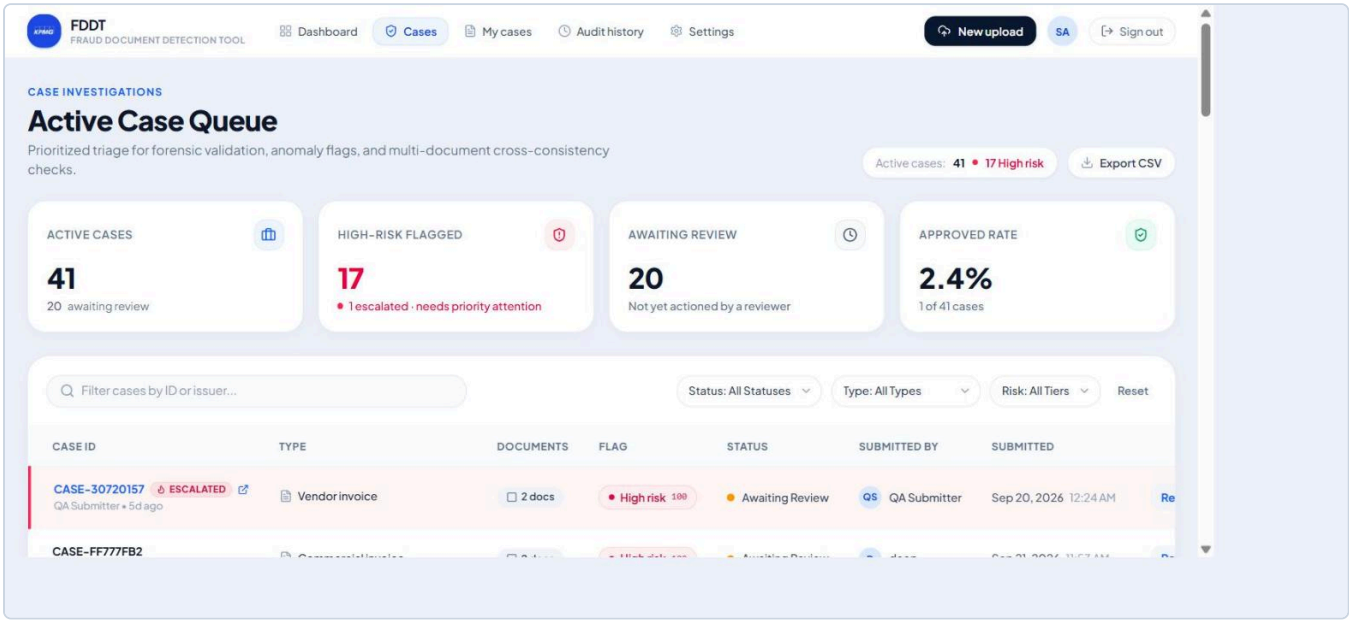
The review queue & dashboard (Reviewers)

Review Queue (also the home page, `/cases`) is the master case list: counts of open / awaiting review / cleared / escalated cases, and a table of cases with their type, status, risk flag, escalation badge, document count and submitter. Search and the status / type / flag filters run instantly in your browser.

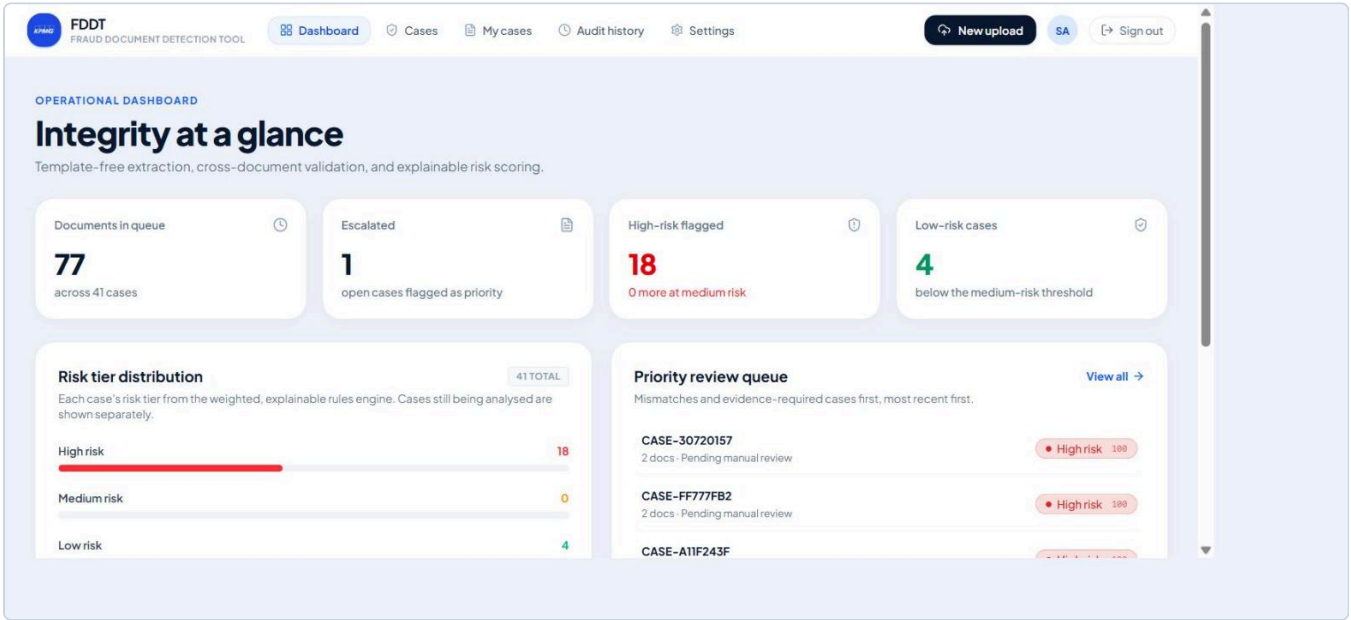
- **Reviewer L1** sees two tabs: **My queue** (the cases you can act on — the default) and **Escalated to L2 · view only** (cases handed to L2; you can open them to follow what happened, but not act on them).
- **Reviewer L2** sees **All cases** (the default — open L2 cases always sort to the top) and **Escalated queue (L2)** (only escalated cases).
- **Platform admins** pick one company at a time (company picker at the top) and see that company's list read-only.

> **"Escalated" is a reviewer tier, not a status.** An escalated case can be `submitted`, `pending_manual_review`, or already decided — escalation means the case now belongs to the Reviewer L2 tier; it doesn't change what state the case is in.

Dashboard is an at-a-glance operational summary computed entirely from the same case list: counts by risk flag, a risk-distribution panel, your top five open cases that you can act on (escalated first, then highest score, then newest), and a summary strip (total cases, documents analyzed, escalated, cleared). It is **not** an organisation-wide compliance/MIS report — there's no fraud-rate trend, average-resolution-time metric, or export here; that class of reporting is out of scope for this build (see *Known limitations*).



The Review Queue: counts, filters, and the case table (admin view)



The Dashboard: KPI cards, risk-tier distribution, and the review queue

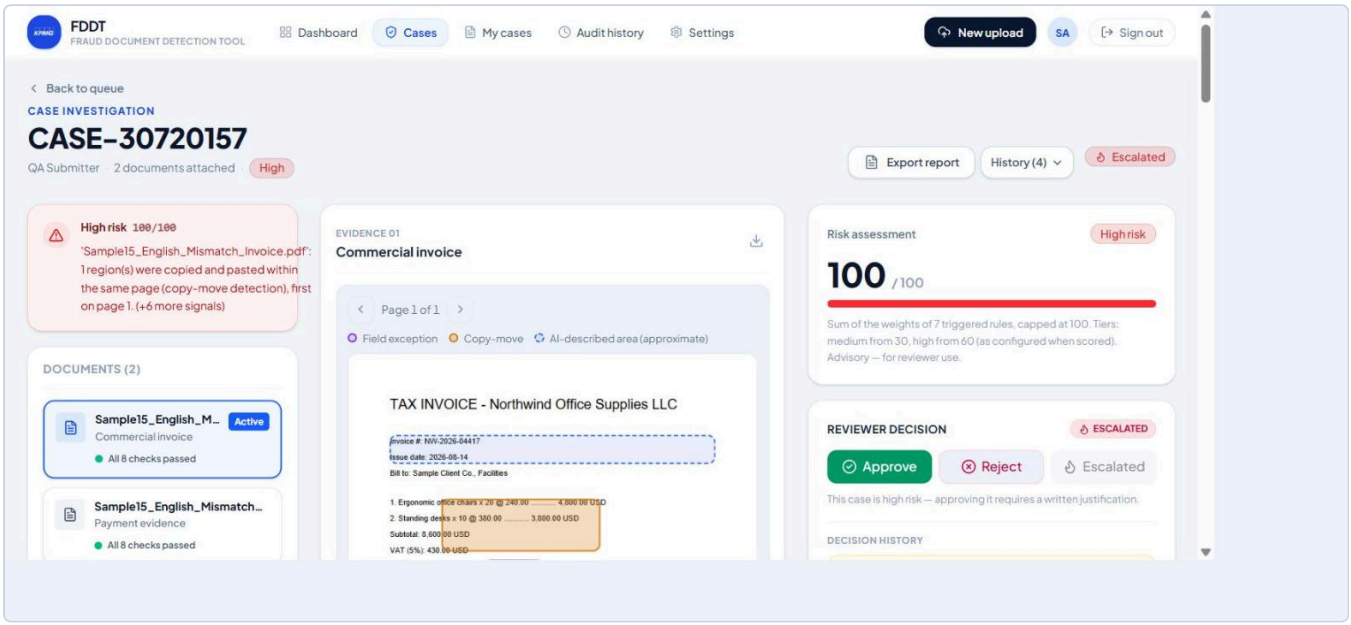
Working a case: the case detail page

Opening a case (`/cases/:caseId`) shows everything about it in three panels plus a timeline:

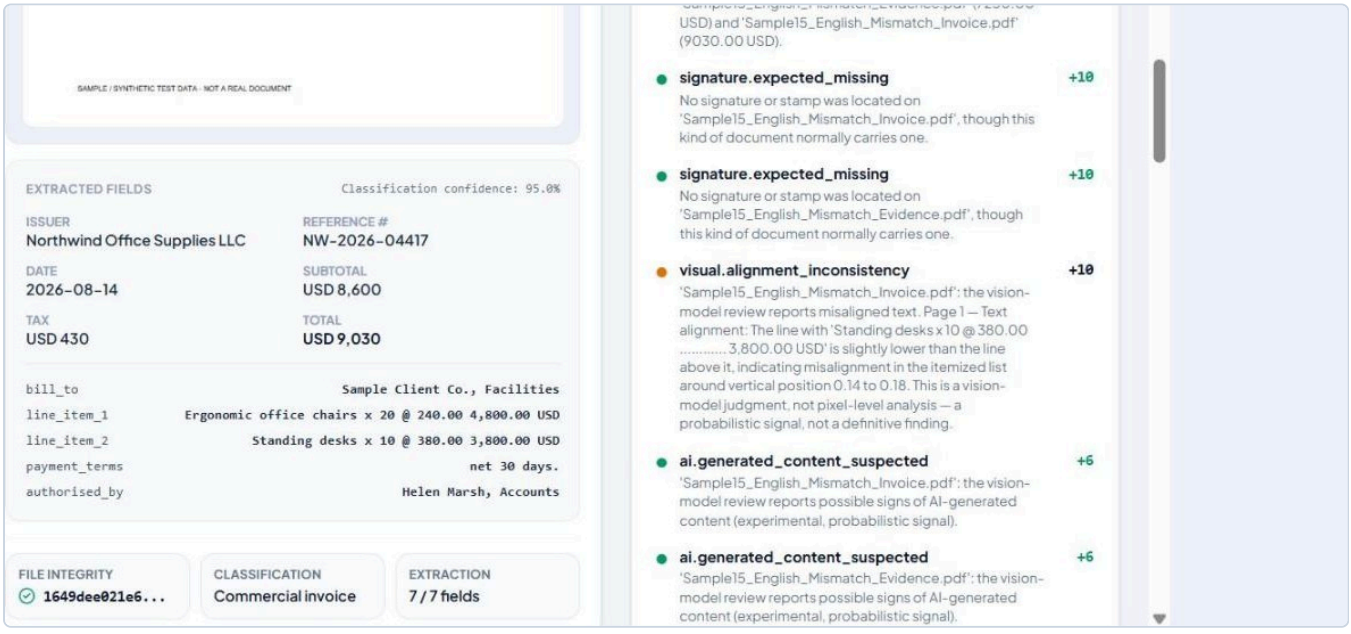
- **Left — Documents:** every file in the case, its type and processing status. Click one to inspect it.

- **Middle — Document viewer & checks:** the selected PDF with live overlays drawn on top of it showing exactly where a check found something; the document's extracted core fields (issuer, reference number, date, amounts — issuer is the only one marked **"(uncertain)"** when the model wasn't confident); and one expandable card per automated check, plus a cross-document card and the signature comparison card if applicable.
- **Right — Risk & decision:** the score, tier, and every triggered reason in plain language; the approve/reject/escalate controls; and (reviewers only) the button to export a forensic report. A platform admin sees an amber **"Support view — read-only"** banner instead of any controls.
- **Bottom — Activity timeline:** the case's full audit trail, newest first — every automated check result and every human action, in order.

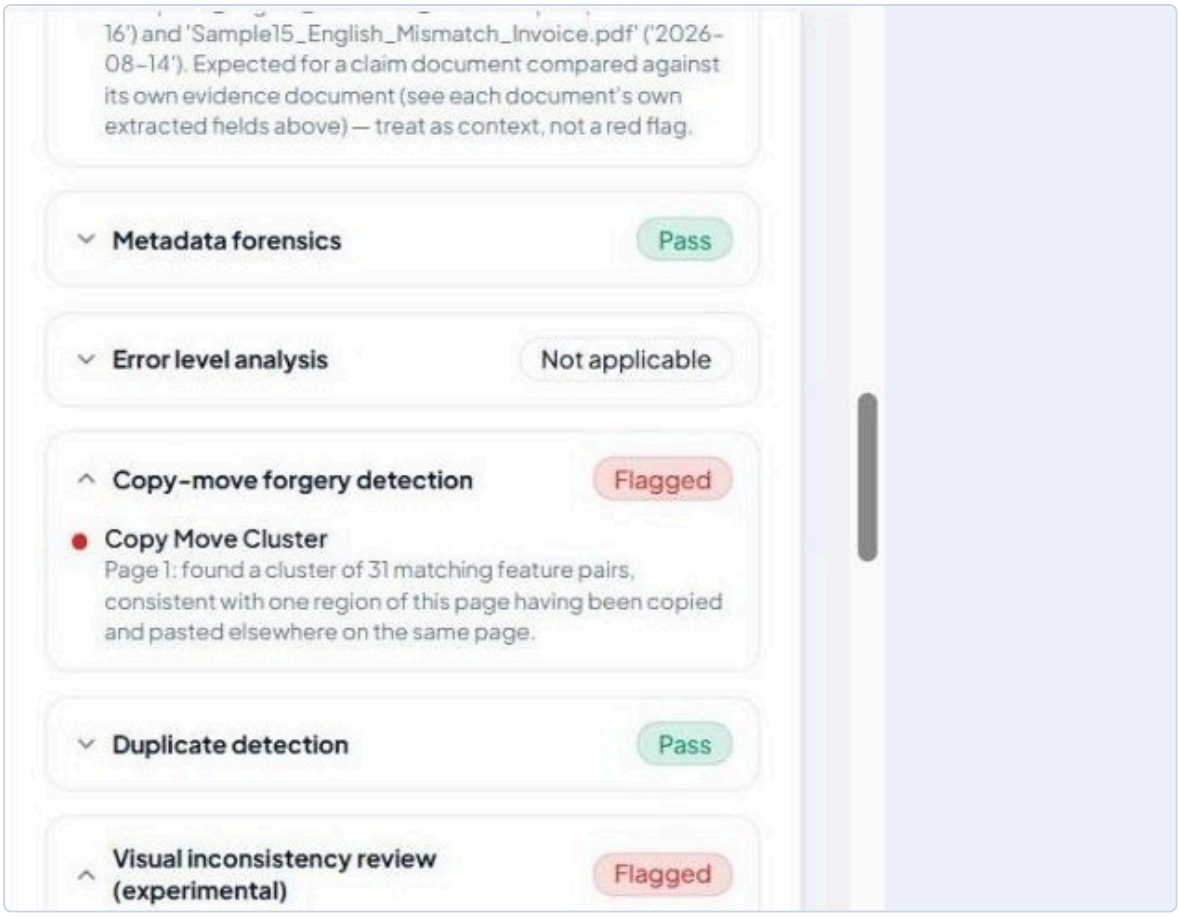
The page **polls automatically every 5 seconds** while the pipeline is still running, so you don't need to refresh — findings and the risk score simply appear as each check finishes.



A high-risk case: documents, the PDF viewer with live overlays, the score, and the decision panel



Extracted fields and the full list of triggered findings in plain language



The per-check cards — each shows its result and, when flagged, the specific finding

Understanding the automated checks

Every uploaded document goes through the same generic pipeline regardless of its type — there's no separate logic per document type, so a school certificate and a vendor invoice get exactly the same scrutiny. In plain language, here's what each check looks for and how far to trust it:

Check	What it looks for	How much to trust it
Extraction & classification	Reads the document (OCR, including printed Arabic), classifies its type, and pulls out issuer, date, amounts, reference number and any other identifiable fields — without a per-type template.	Low-confidence fields are marked; the case still goes to manual review either way.
Field validation	Rule-based checks on the extracted data: is the date in the future, does the total match the sum of line items, are required fields present.	Deterministic and reliable — it's arithmetic and date logic, not a model judgment.
Issuer verification	Fuzzy-matches the extracted issuer name against the known-issuer registry (vendors, schools, tax IDs the organisation has already vetted).	Matches on **name only** — there's no letterhead/logo comparison, and a legitimate new issuer will simply not be in the registry yet.
Cross-document consistency	Once a case has 2+ documents, compares shared fields (amount, date, vendor) pairwise across them and flags mismatches.	Deterministic comparison of extracted values; only as good as the extraction behind it.
Metadata forensics *(PDF only)*	Inspects the PDF's own internal structure: a modify date long after the create date, editing-software fingerprints (e.g., Photoshop) in the tool history, stripped metadata, multiple incremental-save markers.	A strong, mostly-deterministic signal, but presence of an edit marker doesn't by itself prove fraud — legitimate re-saves happen.

Check	What it looks for	How much to trust it
Error Level Analysis (ELA)	Looks for recompression differences that can indicate a pasted-in or edited region of an image.	**A signal, not a verdict.** Weak against a printed-and-rescanned tamper, and can false-positive on a legitimately recompressed scan — weighted lightly in scoring for this reason.
Copy-move detection	Looks for a region duplicated within the same image (e.g., a copied stamp or number).	Classical computer-vision matching — solid at finding literal copy-paste, not a semantic read of what changed.
Duplicate / near-duplicate detection	Perceptual-hashes every uploaded page and compares it against everything your company ever uploaded, including other cases.	Reliable for catching the same file (or a lightly re-saved copy of it) resubmitted — including across unrelated cases. It never compares against another company's documents.
Visual review & AI-generated-content assessment	An AI vision pass describing visual inconsistencies as a secondary check, plus one experimental question about whether the image looks AI-generated.	**Explicitly experimental** — real-world accuracy on scanned business documents is unproven. Treated as one weak signal among many, never a standalone verdict.
Signature / stamp detection & comparison	Locates a signature/stamp region on the page, and — only if a reference signature was set — gives an advisory visual comparison ("visually consistent with reference on file") against it.	**Presence/placement confirmation for a human, not an identity match.** This is a known weak point; the wording deliberately never says "verified".

You will not see a single automated "fraud / not fraud" verdict anywhere in the product — every finding above feeds the risk score as one input among several, and a human always makes the final call.

Understanding the risk score

Every case is scored by a **transparent, weighted rules engine** — not a black-box model. Every point in the score traces back to a named rule and a plain-English reason a reviewer can read.

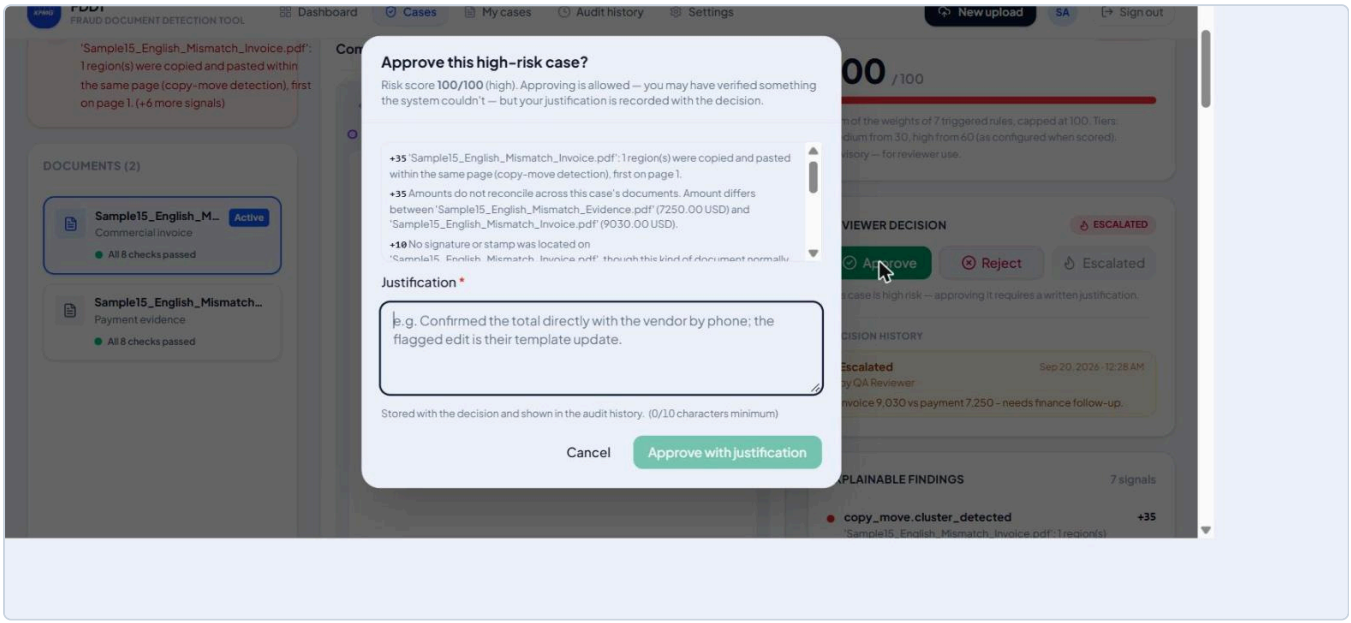
- Each rule that fires adds its configured weight; the total is capped at 100.
- The score maps to a tier: **Low** (0–29 by default), **Medium** (30–59), **High** (60–100) — your company's Reviewer L2s can retune these thresholds under Settings.
- Next to the score you'll see every **triggered reason** in plain language (e.g. "*Issuer name 'X' did not match any known vendor in the registry*"), not just a number.
- **Every scored case goes to the reviewer queue.** Nothing auto-approves, regardless of score — a low score means "no rules fired," not "cleared."
- A case can re-score itself while you're looking at it (for example, when a signature comparison finishes after you opened the case) — a small delay before the numbers settle is expected.
- **When a rule's weight is later retuned, past cases keep the score they were originally given** — the rule versions and thresholds active at scoring time are frozen with that case forever, so tuning the engine never silently rewrites history.

Making a decision: approve / reject / escalate

From the case detail page, a reviewer can do the following. On a case escalated to L2, only a Reviewer L2 can; a Reviewer L1 sees a read-only panel explaining that the case is with L2, and the backend refuses the actions too. Platform admins can never act on a case.

- **Approve.** Disabled until every automated check has finished and the case has a risk assessment (a list of what's still pending is shown). Approving a **low-risk** case is a single click. Approving anything **medium or high** opens a confirmation showing the score and every triggered reason, and requires a **written justification of at least 10 characters** — you may have verified something the system couldn't, but that reasoning is recorded with the decision and shown in the audit history.
- **Reject.** Always requires a reason. It's stored, shown in the audit trail, and visible to the submitter.
- **Escalate.** Requires a reason. This hands the case to the **Reviewer L2** tier — it does **not** change the case's status. Escalation is one-way: there is no "send back to L1". A Reviewer L2 resolves the case with the ordinary Approve / Reject. From then on, Reviewer L1s see the case read-only.

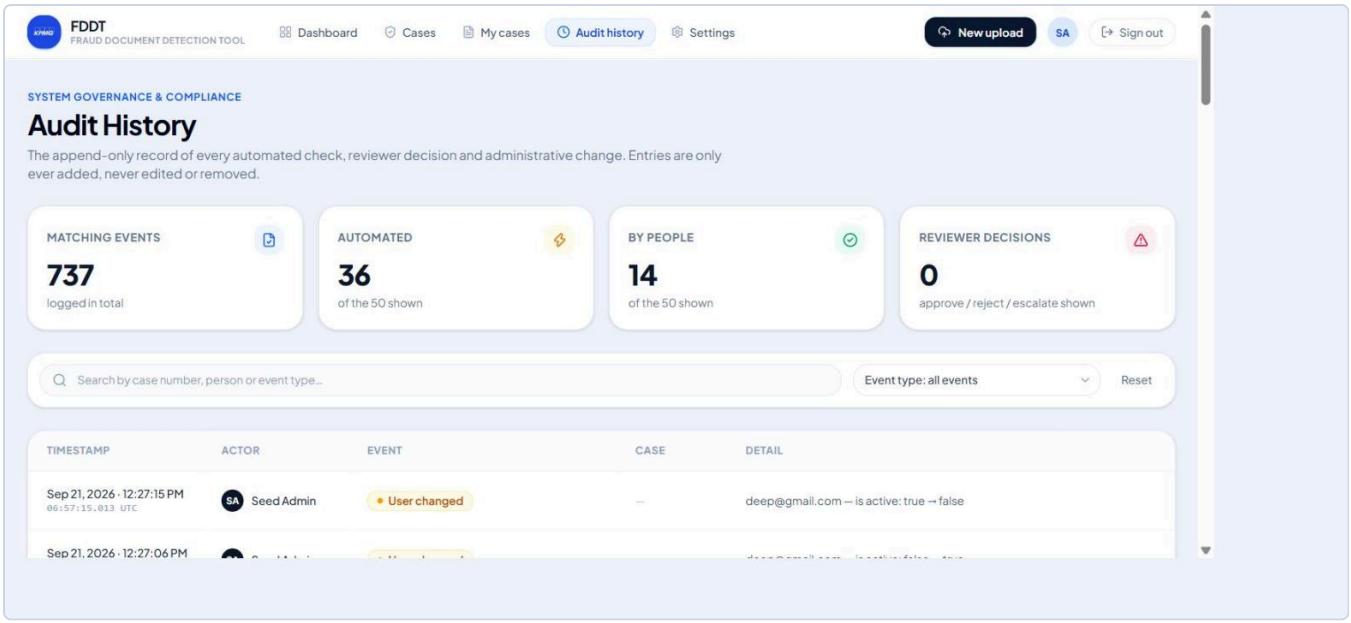
Once a case is approved or rejected, the decision panel becomes read-only.



Approving a high-risk case requires a written justification — the triggered reasons are shown alongside

Audit history

Audit History (`/audit-history` , reviewers only) is a searchable, filterable read of your company's **append-only** audit log — every automated check result and every human action, across every case of the company, written in plain sentences. Nothing in this log is ever edited or deleted; a case's own timeline (on its detail page) is the same log filtered to that case. Platform admins can read a company's log (and a separate platform-level log); their own support views are recorded only in the platform log.



The system-wide, append-only audit log with filters and search

Exporting the forensic report

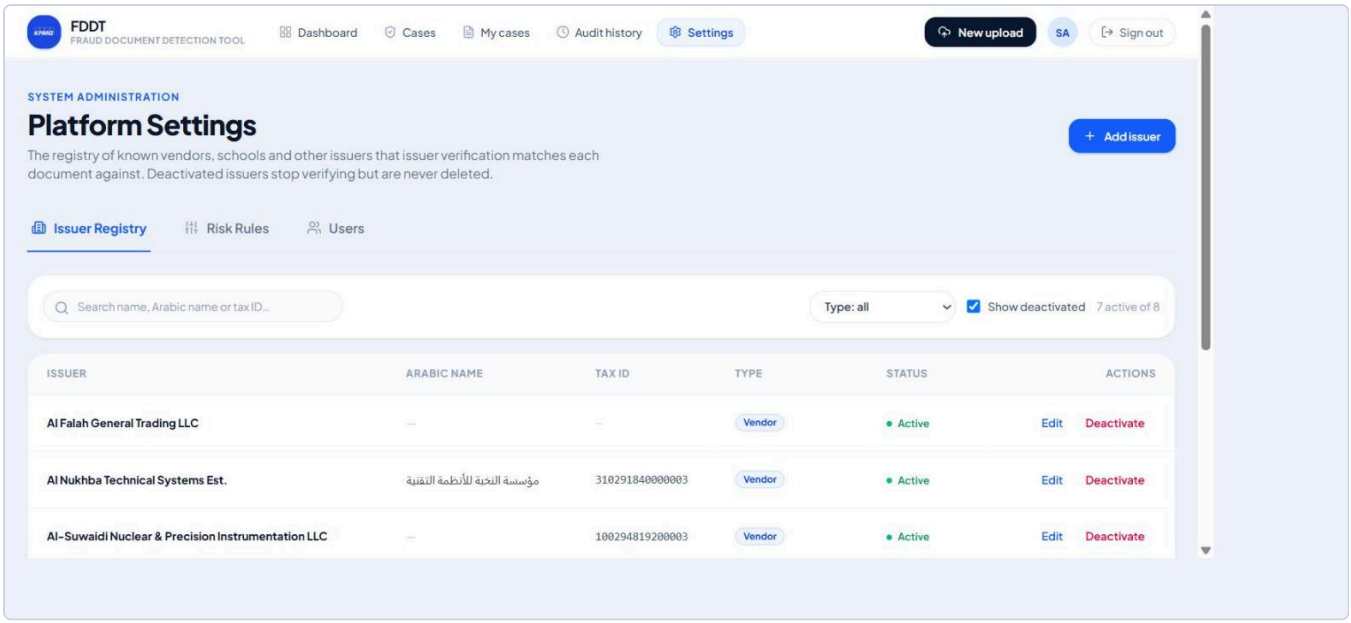
From a case's detail page, a reviewer can generate a standalone PDF report — **Export report** — built to be understood by someone who has never used the application. It contains:

1. Case details (number, type, status, submitter, documents, decision).
2. Executive summary (score, tier, decision, pointers to the sections below).
3. Every triggered risk reason, with severity and the exact region on the page it refers to.
4. The complete checklist — every check on every document, plus the cross-document and signature results.
5. Exceptions — everything that did **not** pass, spelled out concretely.
6. Limitations & assumptions — what the report is (a system-assisted analysis to support human review, not a certified forensic opinion), which checks are deterministic versus model judgment, and which checks actually ran.
7. The full audit trail (oldest first; later events beyond the first 150 are counted, not dropped).
8. An appendix: extracted fields with confidence, each original file's SHA-256 with live tamper-evidence verification, and technical parameters of each check.
9. One annotated page per flagged document page, with every finding highlighted and a legend.

Generating a report takes a few seconds and never re-runs a check or changes the score — it's a frozen, point-in-time snapshot. Every generation is kept; earlier reports for the same case remain downloadable, and two reports for the same case can legitimately show different numbers if the case was re-scored in between.

Administration: issuer registry (Reviewer L2)

Settings → **Issuer Registry** maintains **your company's** list of known vendors, schools and other issuers (name, Arabic name, tax ID, type, active/inactive) that the issuer-verification check matches against. A new company starts with an empty registry — until it is filled in, almost every document will be flagged "issuer not in registry". Deactivating an issuer keeps its history — rows are never deleted, only marked inactive.



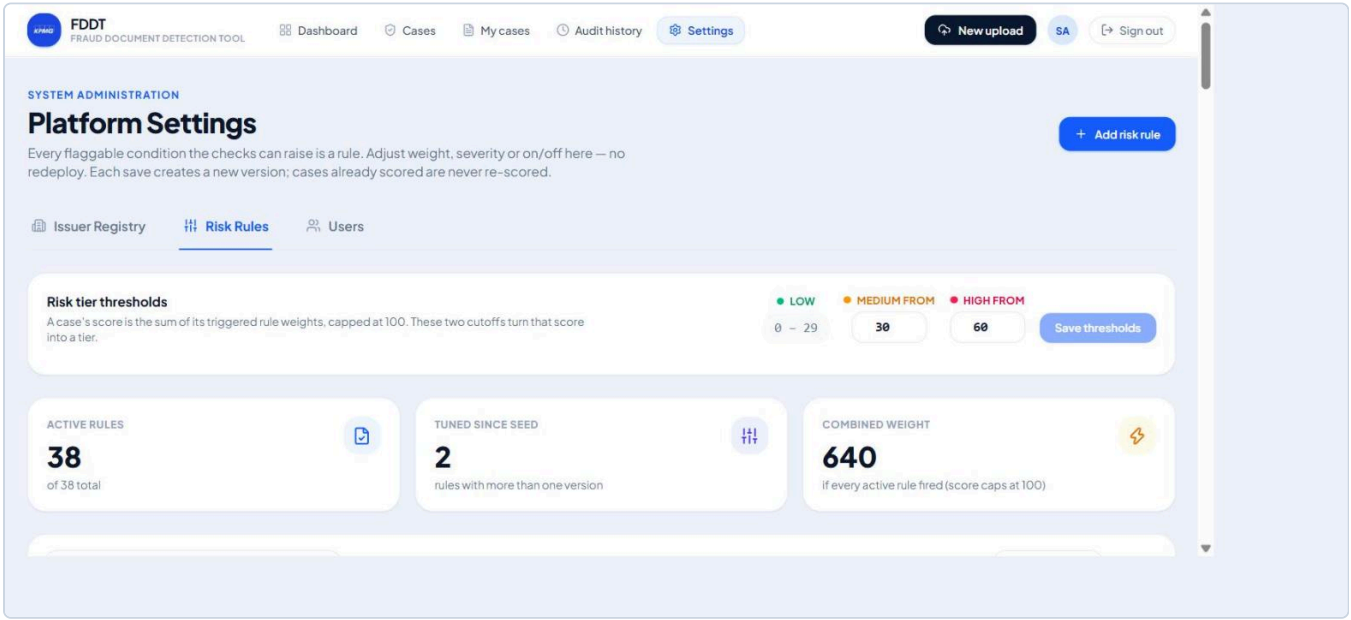
The issuer registry, including right-to-left Arabic names

Administration: risk rules & thresholds (Reviewer L2)

Settings → **Risk Rules** lets you view and tune **your company's** scoring engine without a redeploy (every company starts from the platform's default rule set and then owns its copy):

- Adjust a rule's **weight**, **severity** or **active** flag. This **inserts a new version** — the old one is kept, and every case already scored keeps the version that was active when it was scored.
- View each rule's change history.
- Adjust the **Low / Medium / High thresholds** (default 30 / 60).
- Add a new rule from the catalog of match types the engine understands — you pick a condition and its parameters from a guided list; nobody hand-writes rule logic, and a combination the engine can't evaluate is rejected outright.

Changes apply to your company's cases scored **from that point on**. They are never applied retroactively, and never affect another company.



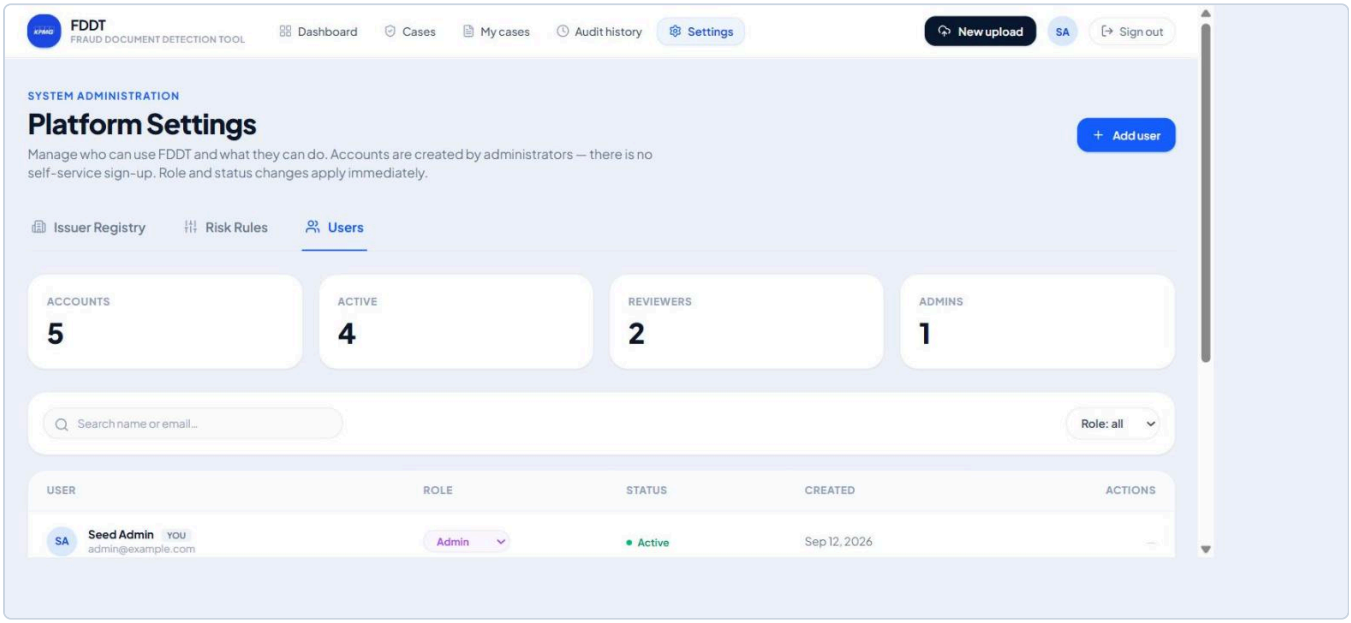
Risk-tier thresholds and the active rule count — tunable without a redeploy

Platform administration (Platform Admin)

The **Platform** area is visible only to platform admins:

- **Companies** — create a company (it receives the default rule set and thresholds; its issuer registry starts empty), rename it, or suspend / reactivate it. Suspending blocks sign-in for all its users.
- **Users** — create and manage every account; there's no self-service sign-up and companies cannot manage their own users. Each account gets a company and a role (User, Reviewer L1, Reviewer L2), or is a Platform Admin with no company. Moving a user to another company signs them out. A platform admin cannot change their own role or deactivate their own account.
- **Rule templates** — the default rules every **new** company starts from. Editing them never changes an existing company's rules.
- **Billing & usage** — per company: cases created, documents uploaded, files stored and storage used, for this month, last month, the last 30 days, this year, all time or a custom range; **Reconcile now** recounts from the source data.
- **Processing queues** — live view of the three processing queues (waiting, running, oldest waiting task, waits, rate-limiter activity, outstanding work per company).

For support, a platform admin can also open any company's case list, cases, audit log and settings (choosing the company at the top of the page). These views are **read-only** and every one is recorded in the platform-only audit log; the company's users don't see those records. A platform admin may change a company's issuer registry or risk rules on its behalf — such a change is recorded in that company's own audit log.



Account management — roles and status, no self-service sign-up

Case statuses & flags — quick reference

What you see	Meaning
Analyzing	The automated pipeline is still running (`submitted` / `under_automated_review`).
Awaiting Review	The pipeline finished; the case is in the reviewer queue (`pending_manual_review`).
Approved	A reviewer approved it.

What you see	Meaning
Rejected	A reviewer rejected it, with a recorded reason.
Escalated · L2 (badge, not a status)	A reviewer handed it to the Reviewer L2 tier; it keeps whatever status it already had.
Low / Medium / High (flag, reviewers only)	The risk tier from the scoring engine. A submitter sees only a neutral "In review" instead.

Known limitations — what this tool does not do

Being direct about this is part of using the tool responsibly:

- **No signature is ever "verified."** Signature/stamp checks confirm presence and placement, and give an advisory visual comparison at best — never a reliable identity match.
- **AI-generated-content detection is experimental.** It's one weak signal inside the visual review, not a proven detector, and its real-world accuracy on scanned business documents hasn't been validated.
- **ELA is a signal, not a verdict** — weak against print-and-rescan tampering, and can flag a legitimately recompressed scan.
- **Only PDFs (up to 10 MB, not password-protected) can be uploaded**, because every forensic check works on the PDF itself. There is no page-count limit and no virus scan.
- **A failed check contributes nothing to the score, silently.** If a check errors out, the case still scores — the failure is visible in the audit log and the check's own card, but no rule fires because of it.
- **The absence of a flagged finding does not guarantee a document is genuine.**
- Not built in this version: notifications/email, SLA tracking and deadlines, organisation-wide MIS/compliance reporting, letterhead/logo matching, cross-case signature-library matching, integration with an external ticketing platform (ServiceNow, Jira, etc.), password reset, and company-managed user accounts.
- **The interface is English-only.** Documents may be in English or printed/typed Arabic; **handwritten Arabic is not reliably read.** Only extracted Arabic field values render right-to-left — the application itself is not translated.

Glossary

Term	Meaning
Company	One client organisation (tenant). Its users, cases, settings and audit log are invisible to every other company.
Case	One submission, grouping one or more related documents (e.g., an invoice plus its payment evidence) under one `case_number` (`CASE-XXXXXXXX`).
Document	One uploaded PDF within a case.
Tier	Low / Medium / High — the risk-score band that drives the case flag and whether approval needs a justification.
Triggered reason	One rule that fired on a case, shown as a plain-language sentence with its severity and weight.
Tier (L1 / L2) / Escalated	Which reviewer tier owns a case. Every case starts at L1; escalating moves it to L2 for good — not a status.
Pipeline	The set of automated Celery jobs (OCR, classification, extraction, validation, forensics, scoring) that run on every uploaded document.
Audit log	The append-only record of every automated result and human action, never edited or deleted.